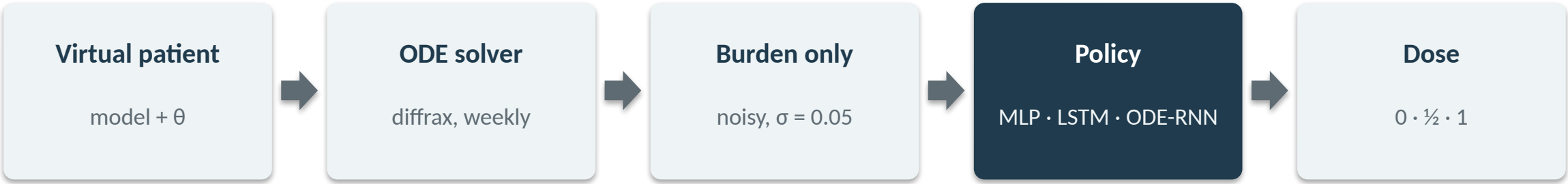
Does a continuous-time policy dose cancer therapy more robustly?

Continuous-time RL for adaptive therapy — what we built, ran, and learned

Program report · 2026-09-27 · virtual patients only, no clinical claims

The question

Is it the differential-equation structure that helps — or just memory?



1 **MLP**

memoryless — the null hypothesis

2 **LSTM**

memory, discrete steps

3 **ODE-RNN**

memory + continuous time between visits

Endpoints, always together: time to progression (TTP: +20% over running nadir, 28-day confirmation) and cumulative drug exposure. Paired by patient, ≥ 3 seeds.

Three tumour models, three different biologies

Lotka–Volterra

competition

$$dS/dt = r_S\, S\, (1 - (S+R)/K)(1 - d_Du) - d_S\, S$$

$$dR/dt = r_R\, R\, (1 - (S+R)/K) - d_R\, R$$

Observed: S + R. Drug slows only sensitive cells; resistant cells are held back by competition — the mechanism adaptive therapy exploits.

Hahnfeldt

angiogenesis

$$dV/dt = -a\, V\, ln(V/K)$$

$$dK/dt = b\, V - d\, V^(2/3)\, K - e\, u\, K$$

Observed: V. The limit is the vascular capacity K, not competition. Drug attacks K.

Kuznetsov

immune predation

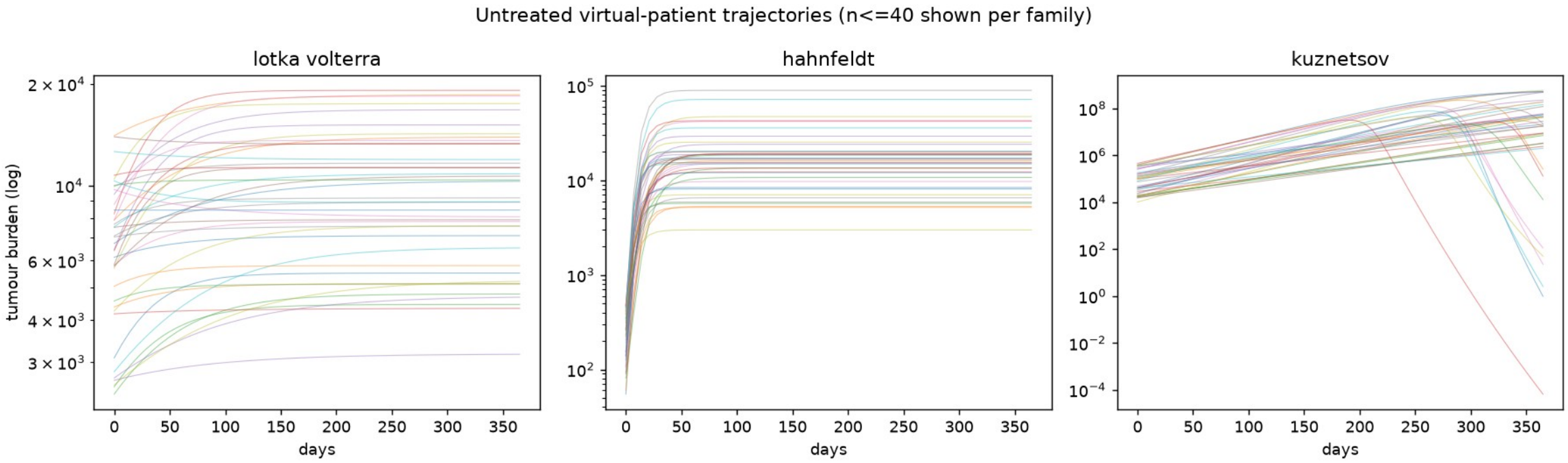
$$dE/dt = s + pET/(g+T) - mET - \delta E$$

$$dT/dt = aT(1 - bT) - nET - uT$$

Observed: T. Effector cells hunt the tumour; oscillation and dormancy are possible. Drug kills T.

Untreated, they behave nothing alike

Phase 0–1: families separable from burden alone (0.996, L48), but θ is not identifiable



Untreated TTP at weekly visits: LV 28 d, Hahnfeldt 7 d, Kuznetsov 14 d (L88). Only LV has dynamic range, so the architecture comparison uses LV.

Dosing regimes compared

Same simulator loop, same weekly visit, only the decision rule changes

none

never treat

MTD

full dose every visit — standard of care

AT50

treat until burden < 50% of baseline, hold until back at baseline (Zhang trial)

thr0.8

treat whenever burden > 0.8 × baseline — the rule Gallagher 2024's RL found

MLP

trained, memoryless

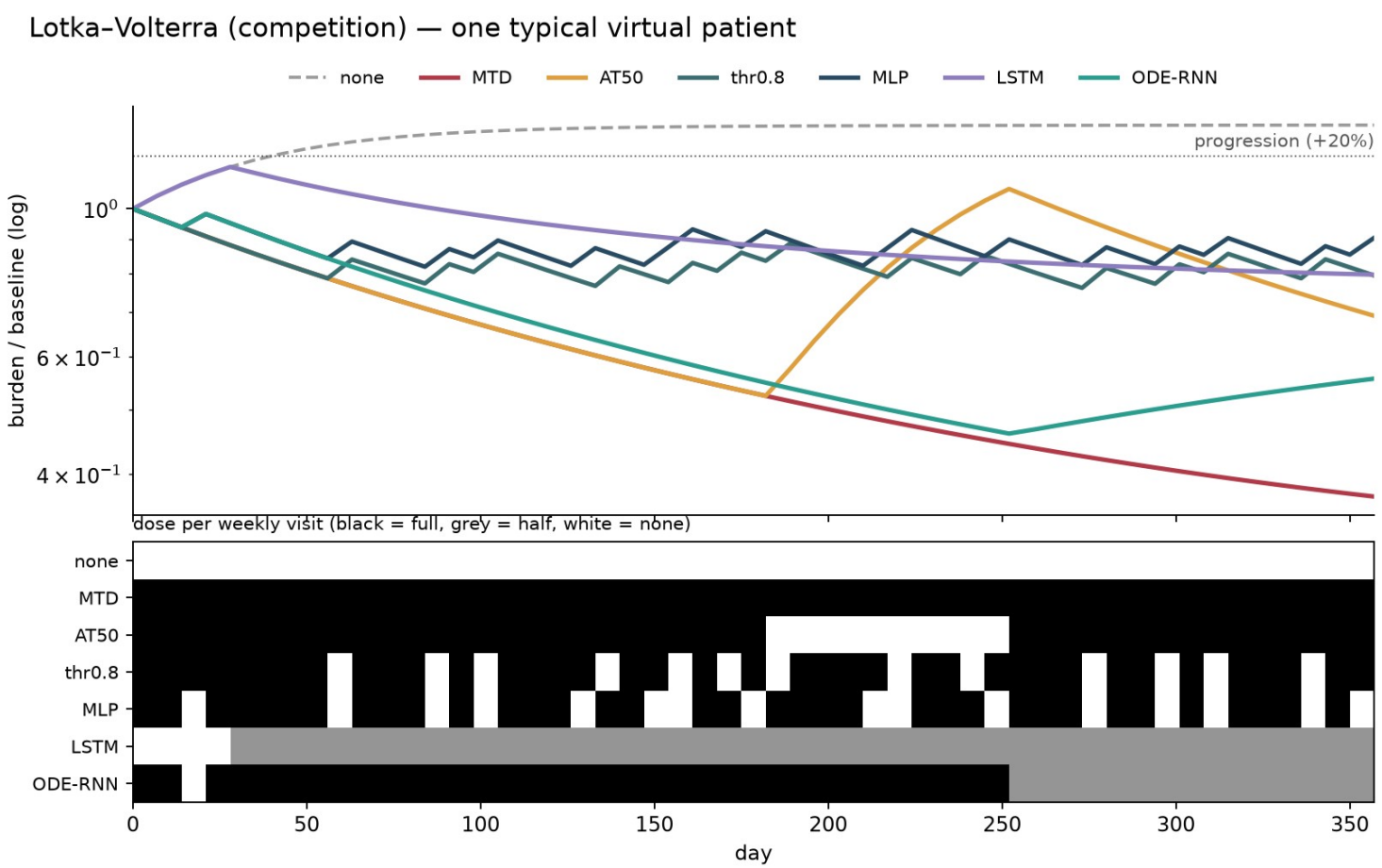
LSTM

trained, discrete-time memory

ODE-RNN

trained, continuous-time memory

LV: holding burden near 80% contains resistance

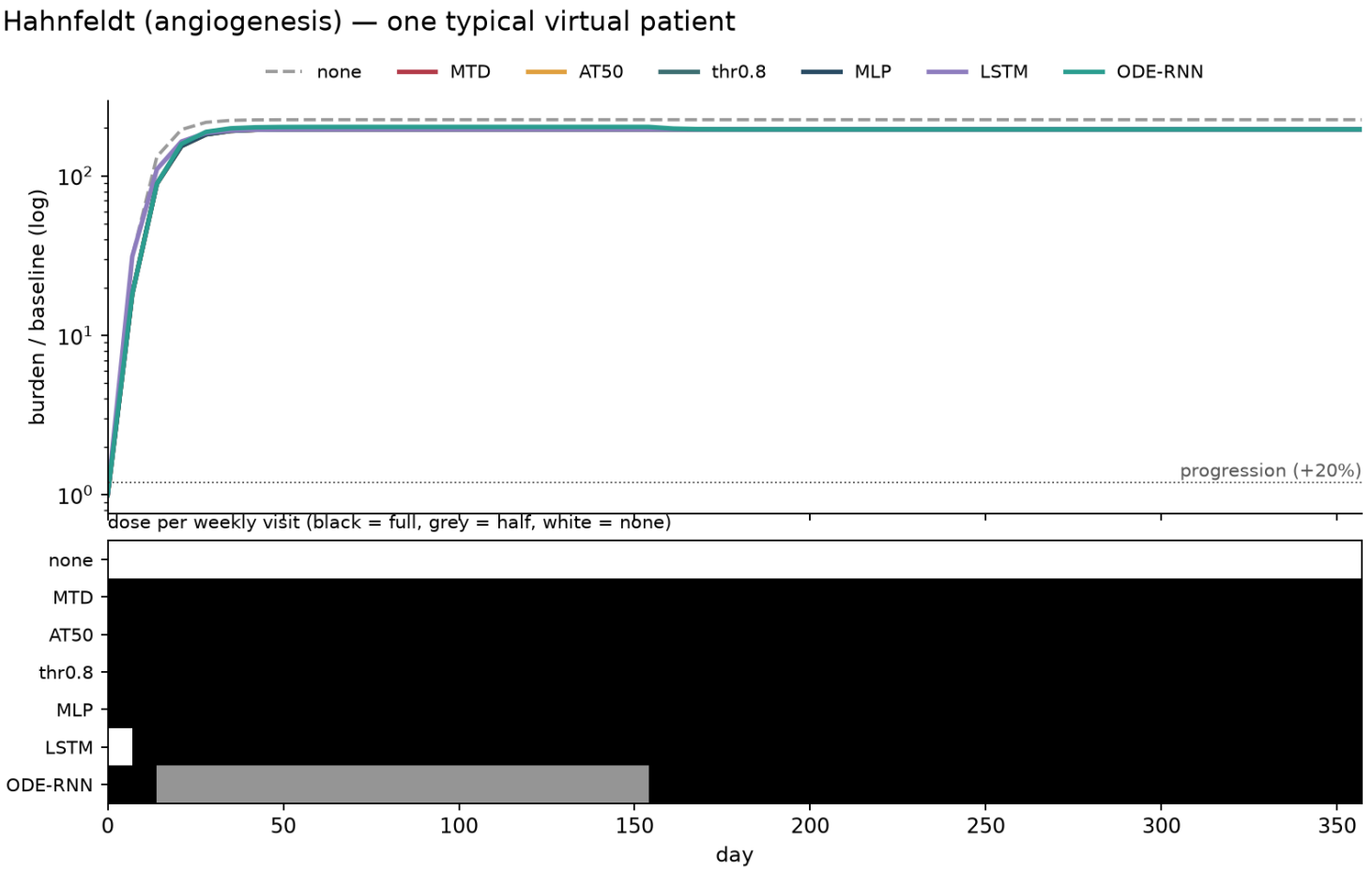


1. MLP and the 0.8 threshold rule cycle the drug and keep the tumour below baseline all year.

2. AT50 lets it rebound to baseline before re-treating; MTD shrinks it fastest with the most drug.

3. The LSTM switches to half dose and drifts; the ODE-RNN holds full dose, then half.

Hahnfeldt: no regime changes the outcome

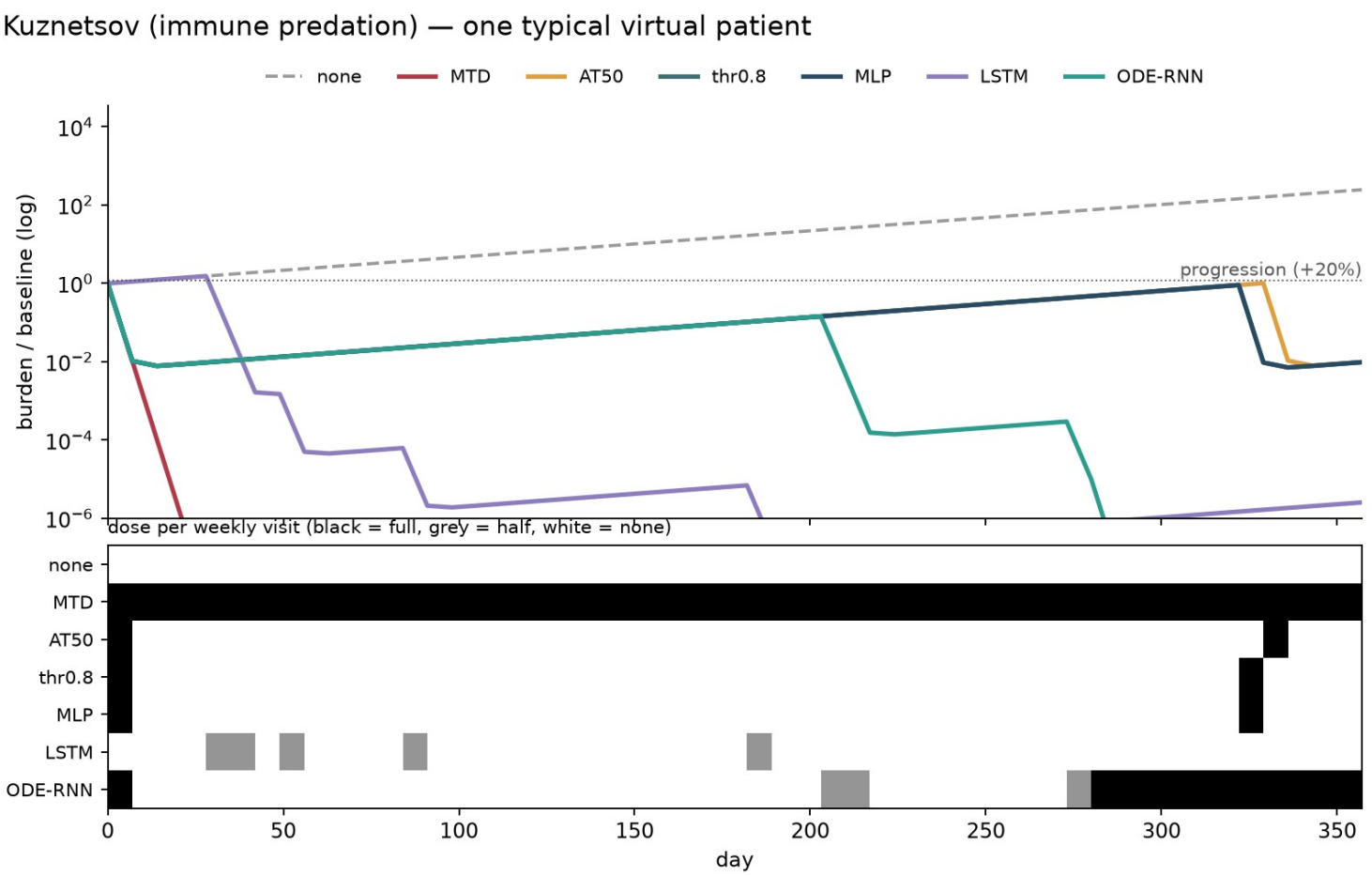


1. The tumour grows ~200× within weeks whatever the dose: every regime progresses on day 7.

2. This is Phase 1's resolution limit (L88): treated TTP has no dynamic range here.

3. W5 (mechanism shift) needs this fixed before it can be registered.

Kuznetsov: only full dose eradicates



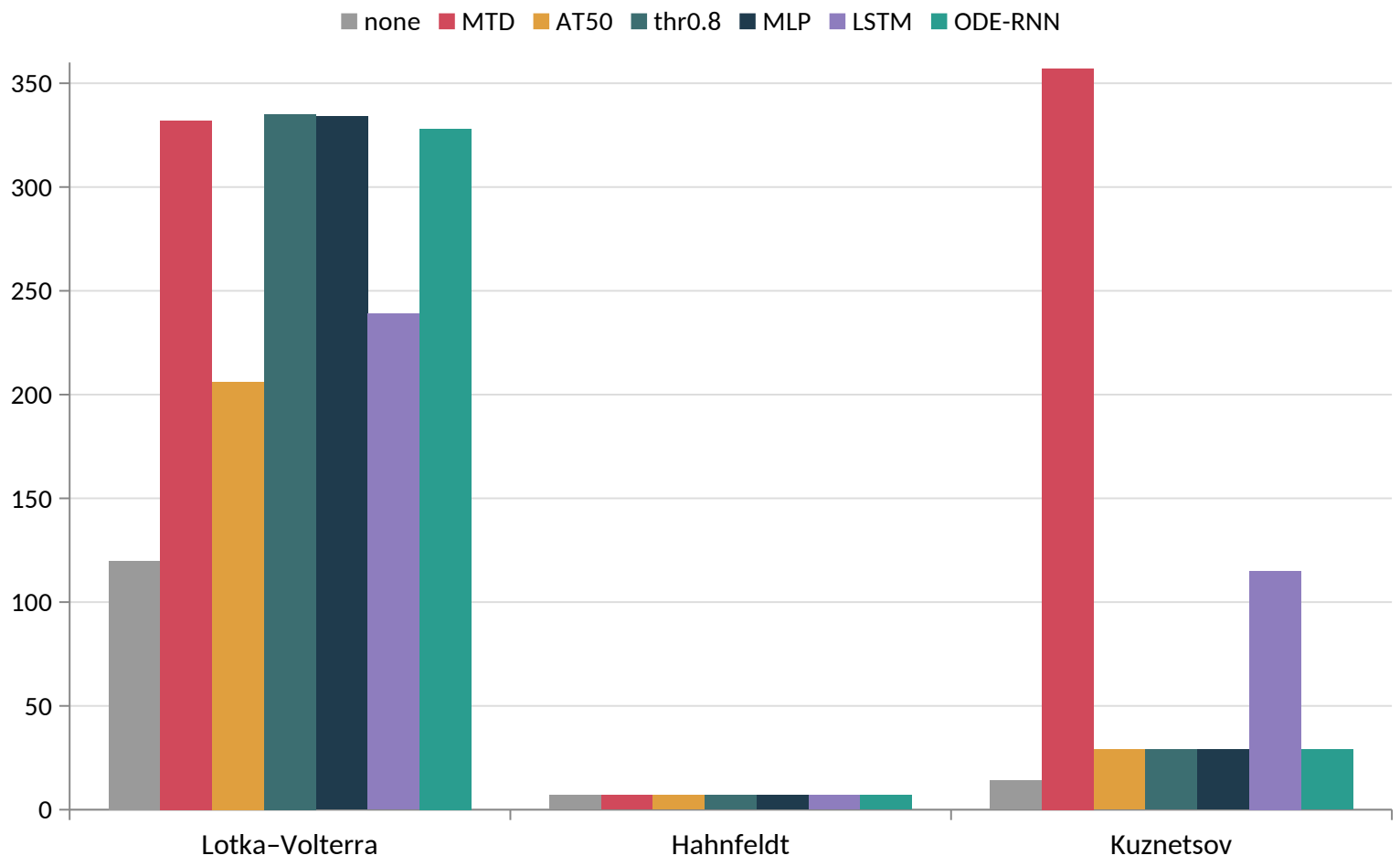
1. LV-style rules treat once, see burden fall 100×, and stop while the immune balance is lost.

2. MTD drives the tumour to extinction; the recurrent nets suppress it at very different drug.

3. A threshold learned on one biology does not transfer to another.

Across 100 patients: TTP bought, and at what dose

Restricted-mean TTP (days, capped at 357) · illustration, not a registered comparison



Lotka–Volterra

MLP 334 d at exposure 245

thr0.8 335 d at 259

MTD 332 d at 357

→ the trained MLP ≈ a one-line rule, with 31% less drug than MTD.

Hahnfeldt

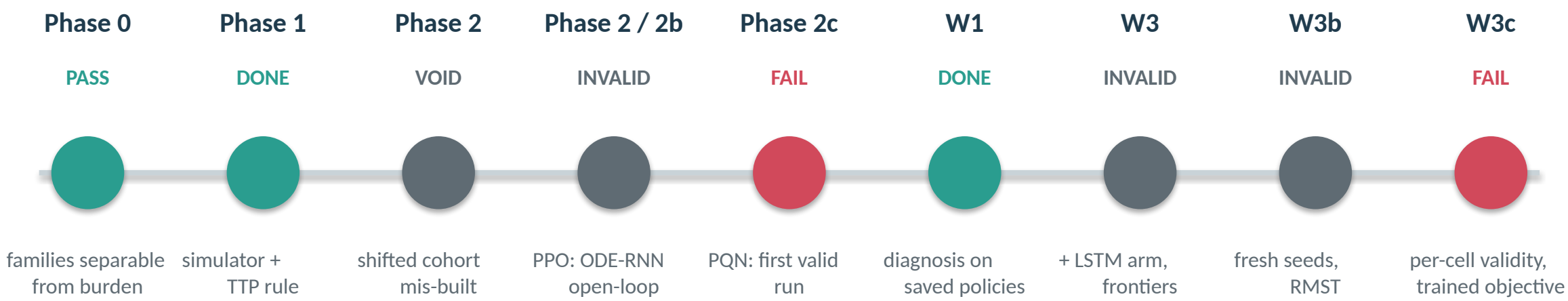
7 d for every regime: no signal at weekly visits.

Kuznetsov

only MTD reaches the horizon (357 d).

What was run, in order

Every comparison pre-registered; void and invalid runs kept in the record



Why so many invalid runs? Each caught a way this evaluation can lie: a policy that ignores its input, a median that rewards letting patients progress, a cohort built from the unshifted parameter, a metric stuck at its ceiling. Each is now a gate in every run.

Phase 2c: the first valid run — the ODE-RNN loses

LV, Tier-1 shift (×1.5 growth rates, ×0.6 capacity), PQN for both arms, 3 seeds × 200 patients

264 vs 19

patients progressing earlier vs later on the ODE-RNN

exposure weight 0.05

337 vs 22

patients progressing earlier vs later on the ODE-RNN

exposure weight 0.20

-31%

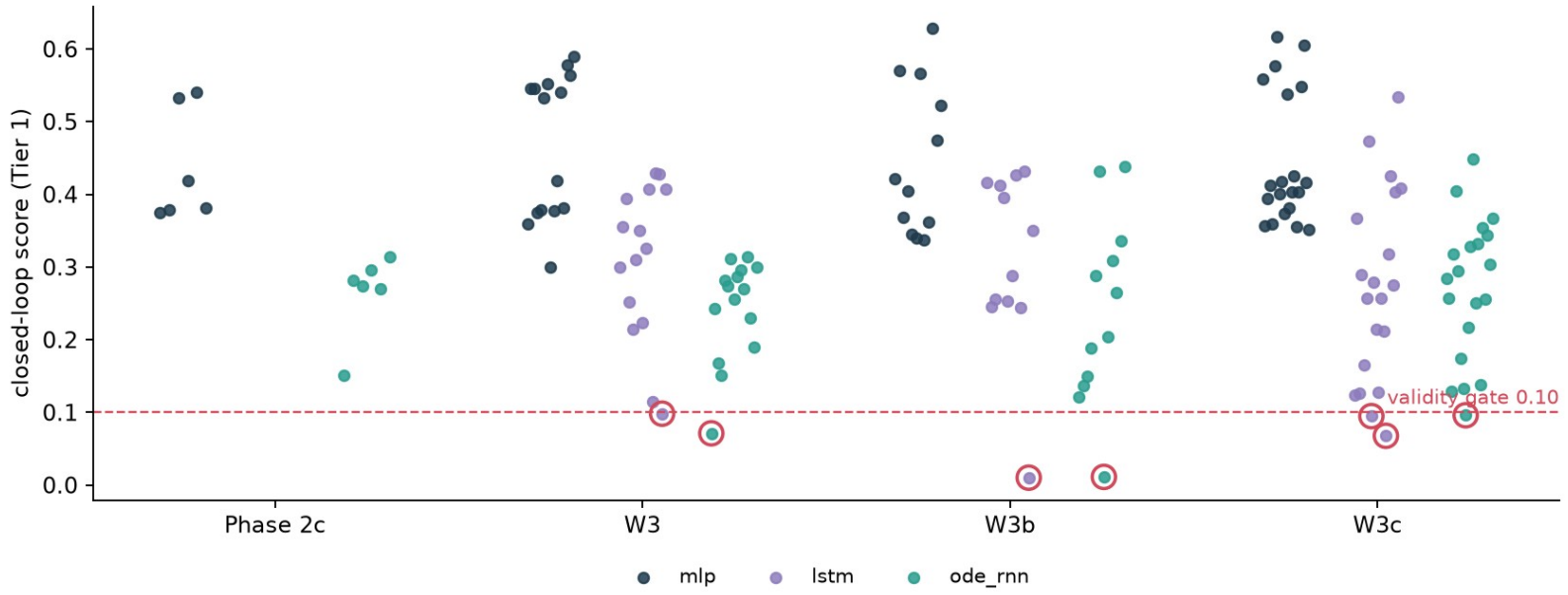
drug given by the ODE-RNN (194 vs 280)

so TTP is not exposure-matched

Kill criterion fired: no continuous-time edge on the clean single-model case, with either trainer (PPO or PQN). The multi-model curriculum was not built.

The recurrent arms sometimes stop listening

Closed-loop score of every trained cell; circled cells fail the 0.10 gate



Below the gate

W3 lstm w0.4 s1: 0.098

W3 ode_rnn w0.0 s0: 0.071

W3B lstm w0.4 s4: 0.010

W3B ode_rnn w0.2 s3: 0.011

W3C lstm w0.1 s7: 0.095

W3C lstm w0.2 s8: 0.068

W3C ode_rnn w0.1 s8: 0.096

The MLP never fails; 7 recurrent cells have. W3c's per-cell rule excludes such cells instead of voiding the arm.

W1: it is a training gap, not a robustness gap

The same policies on the patients before the shift (Tier 0)

Tier 0 (no shift)

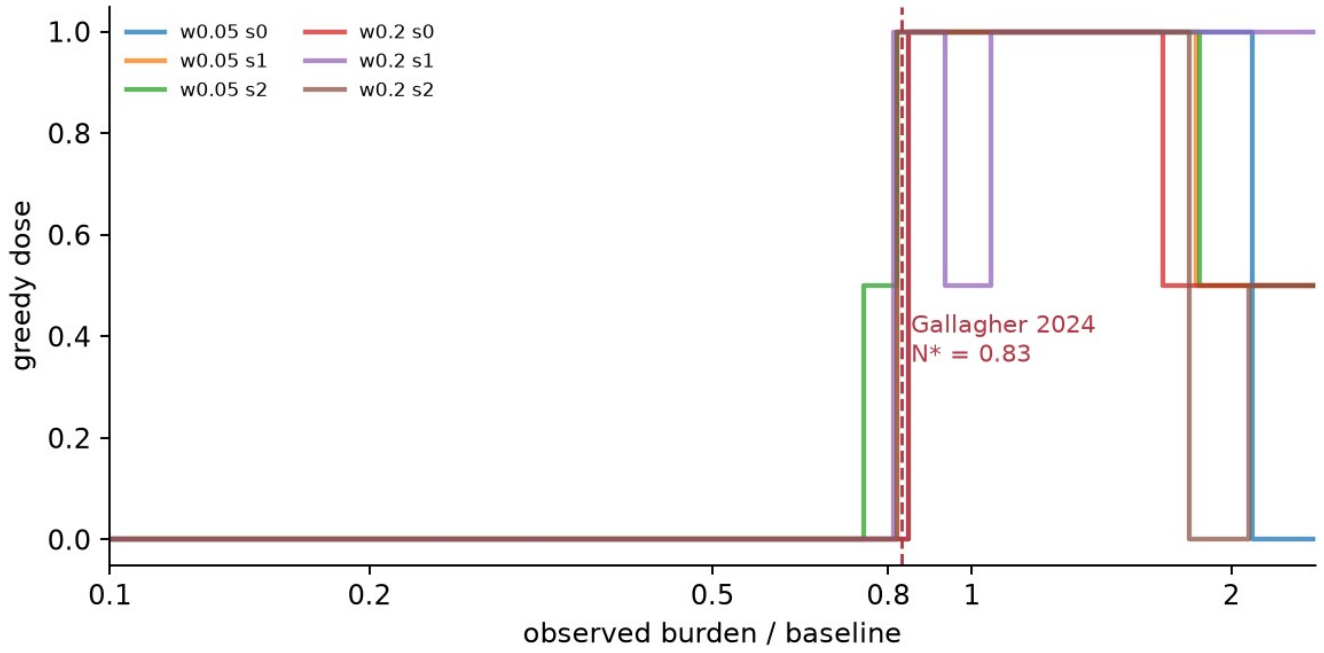
164 vs 10

ODE-RNN earlier vs later (w 0.05)

Tier 1 (shifted)

264 vs 19

the shift moves both arms, not the gap



The MLP learned a switch: no drug below ~0.8 × baseline, full dose above, in all 6 seeds (≥ 94% of doses explained by the current reading). The ODE-RNN doses partly from memory (64%–88%).

What the two closest papers actually show

Read in full from the PDFs, with page locators

Gallagher et al. 2024, Cancer Research

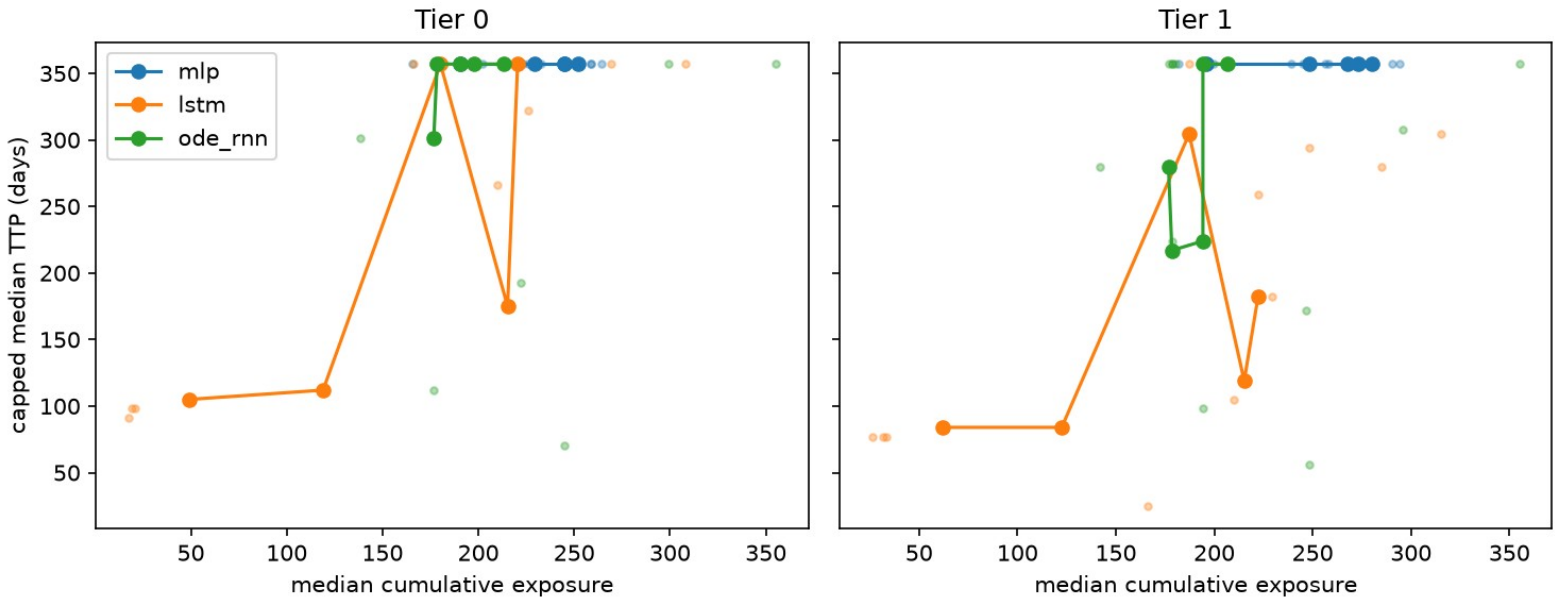
- Same setting: LV, burden-only, TTP = +20%.
- Their RL policy is a switch at $N^* = 0.83 \times n_0$ (Fig. 4B, p.1935) — our MLP found ~0.80.
- Evaluating less often than trained lowers TTP (Fig. 3C, p.1934): predicts W4's direction.
- Means over 100 evaluations; exposure not matched.

Kiram et al. 2026, arXiv 2605.02552

- Recurrent TD3 380 ± 12 vs TD3 −50 ± 252 (Table I, p.5).
- But: one fixed patient, no shift, means over 10 seeds.
- The hidden variable is a toxicity state in the reward — not our S/R split.
- Justifies an LSTM arm; not evidence that memory helps here.

W3: adding an LSTM arm

MLP vs LSTM isolates memory; LSTM vs ODE-RNN isolates continuous time



Memory: LSTM − MLP

Tier 1 -182 d [-233, -136]

overlap 16% · INVALID

Continuous time: ODE − LSTM

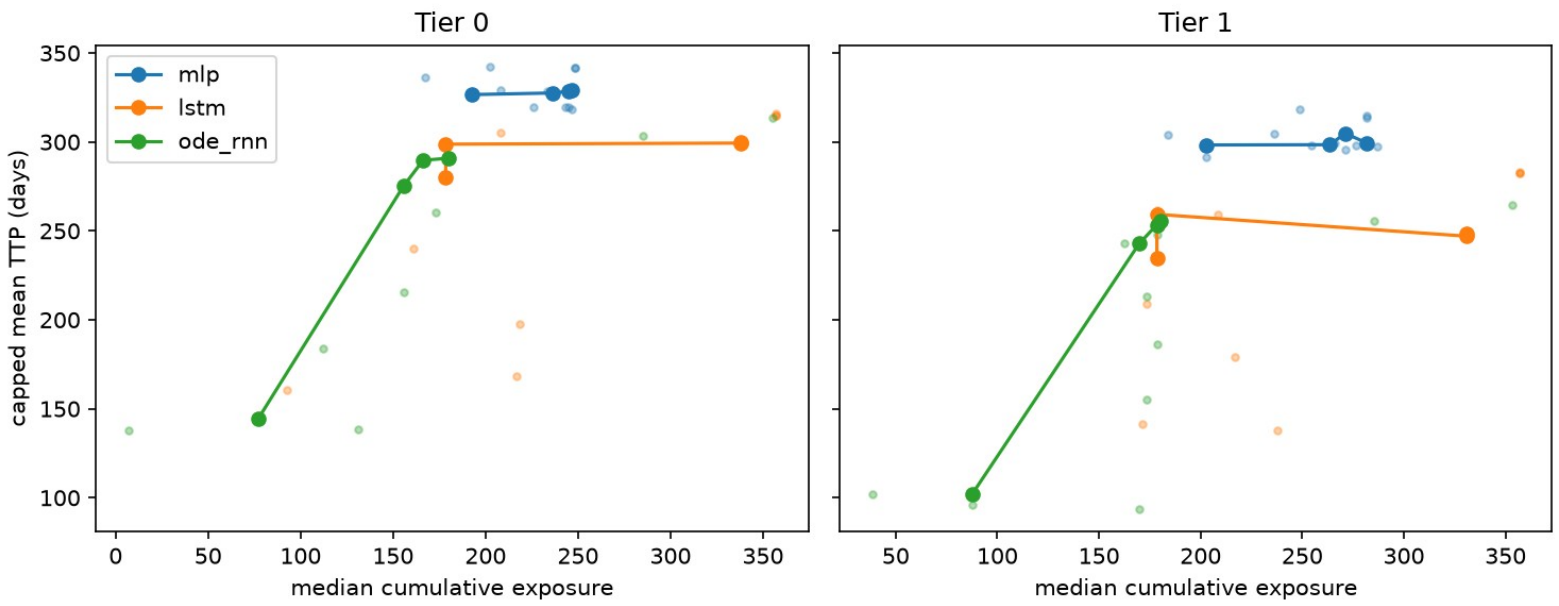
Tier 1 +22 d [-41, +158]

overlap 19% · INVALID

No verdict: arms with an open-loop cell: lstm, ode_rnn. Gaps are descriptive only.

W3b: fresh seeds, restricted-mean TTP

Same design; the ceiling removed



Memory: LSTM − MLP

Tier 1 -45 d [-59, -40]

overlap 52% · INVALID

Continuous time: ODE − LSTM

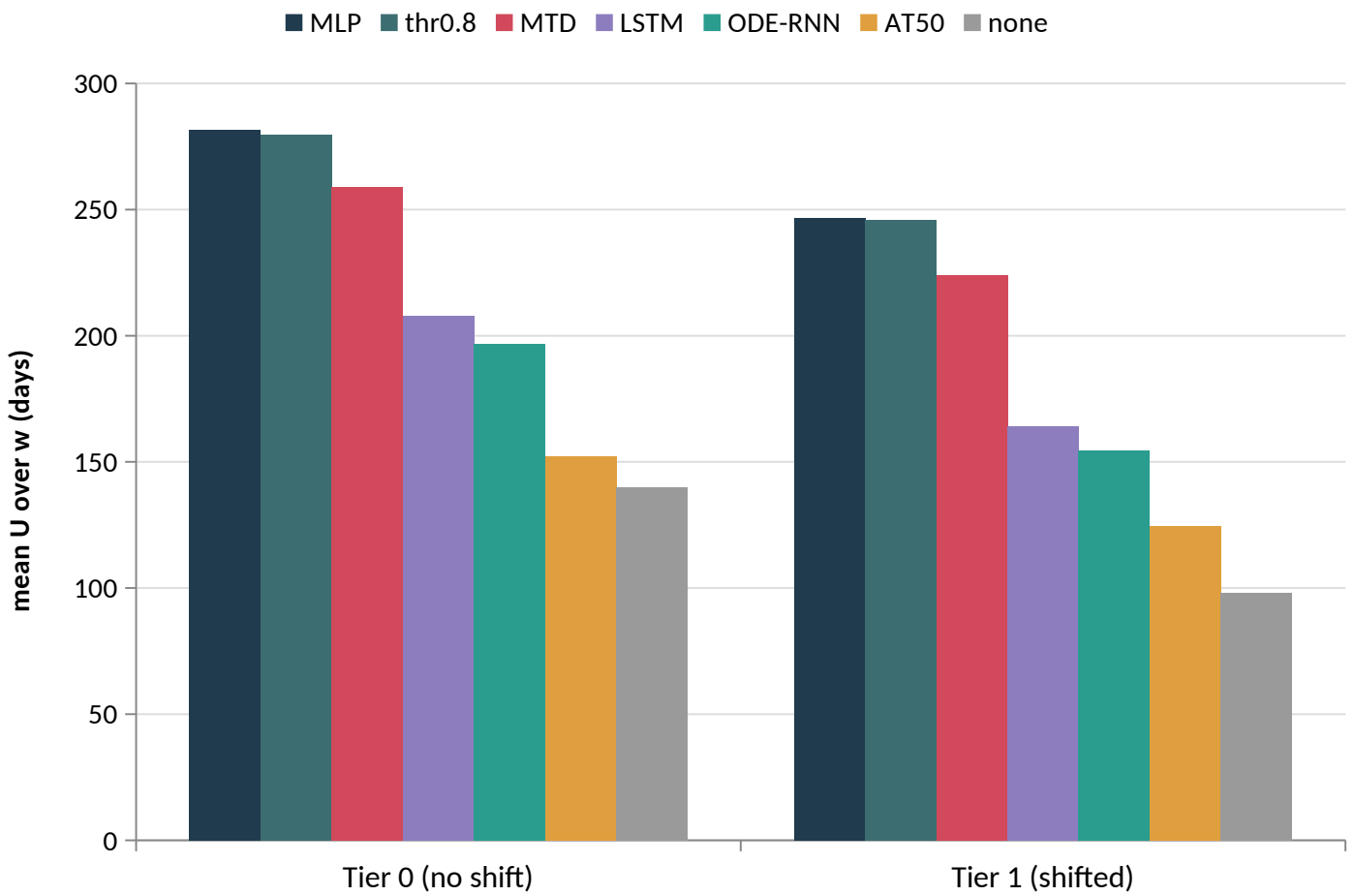
Tier 1 -5 d [-15, +12]

overlap 1% · INVALID

No verdict: arms with an open-loop cell: lstm, ode_rnn. Gaps are descriptive only.

W3c: memory hurts, continuous time hurts further

First valid three-arm run · trained objective U = TTP − w · exposure, paired per patient



Memory: LSTM − MLP

-82.9 d [-88, -78] Tier 1

Tier 0: -73.5 d · FAILS

Continuous time: ODE − LSTM

-15.7 d [-20, -11] Tier 1

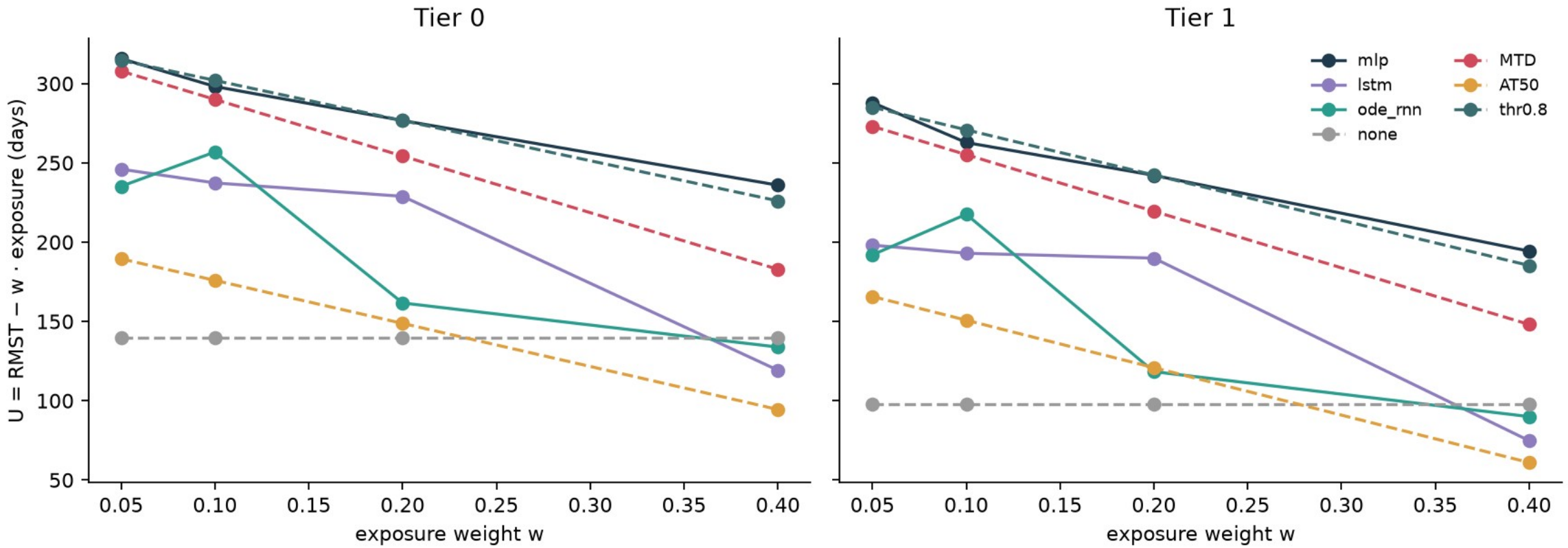
Tier 0: -16.7 d · FAILS

Excluded cells: 3/60, none from the MLP

The MLP only ties the one-line 0.8 × baseline rule.

W3c per exposure weight

Dashed = fixed rules, solid = trained policies



The gap is present without the shift (Tier 0), so it is a controller-quality gap, not a robustness gap — the same shape W1 found for Phase 2c.

Four ways this evaluation lied — each now a gate

1 An open-loop schedule

The ODE-RNN gave all 200 patients one identical dosing history and still reported a respectable TTP (L312).

Fix: closed-loop validity ≥ 0.10 per cell

2 The progressor-only median

Averaging only patients who progress rewards letting more of them progress; it inverted the paired result 3 times.

Fix: min(TTP, horizon) over all patients

3 A mis-built shifted cohort

Scaling K after building y0 put half the test cohort above capacity; NaN blow-ups scored as 'never progressed' (L242).

Fix: rebuild y0 from shifted θ; refuse NaN

4 A metric at its ceiling

Once > 50% never progress, a capped median is flat at the horizon and ranks nothing (L496).

Fix: restricted-mean TTP (W3b)

Next

Target: ICLR 2027 workshop — a clean negative is still the paper

Law promote 'no continuous-time edge on Tiers 0/1' — two valid closures (vs MLP, vs LSTM)

W4 irregular visits (20% / 50% dropped) — where continuous time has a mechanical reason to help; LSTM + Δt as the strong discrete baseline

W5 mechanism shift — only if Hahnfeldt's day-7 resolution limit is fixed

W6 write-up: validity across trainers, frontiers, the learned threshold rule, every run in one table